

MUHAMMAD SHAH

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Education

FAST-NUCES (National University of Computer and Emerging Sciences) Islamabad, Pakistan
B.S. in Software Engineering 2022 – 2026

- **Relevant Courses:** Artificial Intelligence , Deep Learning , Computer Vision (A Grade), Machine Learning, Professional Practices in IT.

Experience

Teaching Assistant – Professional Practices in IT FAST University, Pakistan
Teaching Assistant – Object Oriented Programming FAST University, Pakistan

Projects

Final Year Project

LexiThera – AI-Powered Speech Therapy Platform | *Node.js, WebRTC, ML, Whisper*

- **Issue Solved & Space Filled:** Solved the critical lack of independent, real-time speech therapy practice outside clinical hours by creating a digitized telehealth ecosystem that bridges the gap between consultants and patients.
- **AI & ML Pipeline:** Designed an advanced AI pipeline utilizing **Whisper AI** for highly accurate speech-to-text inference. Implemented dynamic audio segmentation and comparative analysis using **MFCC feature extraction** and **Dynamic Time Warping (DTW)** to evaluate acoustic signals.
- **Impact:** Provided patients with instant, granular, word-level pronunciation scoring for asynchronous practice, while also supporting live WebRTC video consultations and secure patient records.

Academic & Independent Projects

Crop Yield Prediction Portal | *Python, Scikit-Learn, Flask, Pandas*

- Built an end-to-end Machine Learning web application predicting crop yields across 101 countries using climate, pesticide, and geographic indicators.
- Engineered a unified preprocessing pipeline utilizing **ColumnTransformer** (StandardScaler and OneHotEncoder) to prevent data leakage, serializing models for deployment.
- Deployed an optimized **Decision Tree Regressor** that achieved a high accuracy **R^2 score of ~ 0.97** , served via a responsive Flask API.

MathMind AI – Learning Platform | *PHP, JavaScript, MySQL, Gemini API*

- Developed a full-stack educational web app featuring 30 levels of timed math challenges with secure user authentication (bcrypt) and dynamic leaderboards.
- Integrated **Google Gemini 2.5 Flash API** as an AI Tutor to generate dynamic, step-by-step explanations when users submit incorrect answers.
- Engineered a robust **offline fallback tutoring engine** to automatically assume control and generate basic explanations if the external API becomes unavailable, ensuring zero downtime.

Technical Skills

Programming Languages: Python, C++, C, PHP, JavaScript, HTML/CSS

Machine Learning & AI: Deep Learning, Computer Vision, Whisper AI, Scikit-Learn, Gemini API, Decision Trees

Audio & Signal Processing: MFCC, Dynamic Time Warping (DTW), Speech Processing

Web Development & Networking: Flask, Node.js, Express, WebRTC, Socket.io, REST APIs

Data Tools & NLP: Pandas, NumPy, Data Cleaning, Visualization

Tools & Platforms: Git, Linux, MySQL, Jupyter Notebook, VS Code, Supabase

Certifications

- **Machine Learning Specialization by Andrew Ng** – *Coursera*